

Homework 01 Solutions

Question 1 (Least squares and matrix calculations)

Solution

R

► Show the solution code

```
[1] 100    6
[1] 6
  coefficient truth  estimate
1 (Intercept)   3.0  3.0063210
2           x1   1.4  1.4171604
3           x2  -0.9 -0.8883453
4           x3   0.7  0.7952834
5           x4   0.0  0.0484378
6           x5   1.1  1.1319341
[1] 0.6763696
[1] 4.52971e-14
```

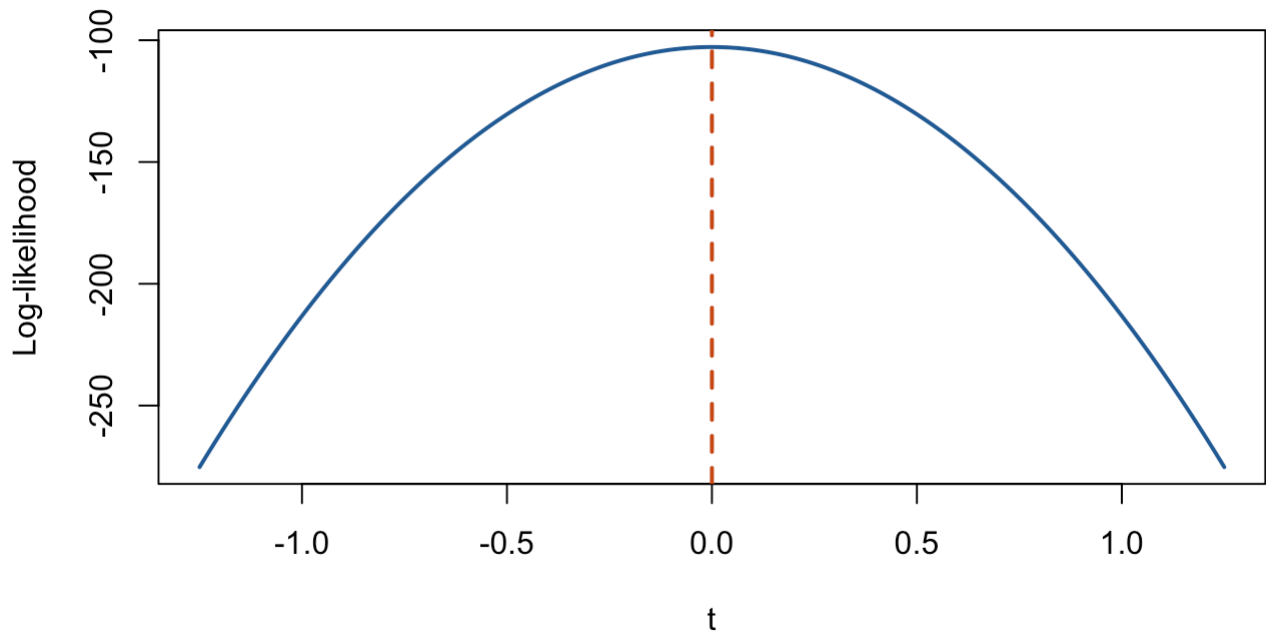
The training RMSE is approximately (0.6764), and $(X^T r_-)$ is about (5.9^{-14}) .

Question 2 (Gaussian likelihood for linear regression)

Solution

R

► Show the solution code



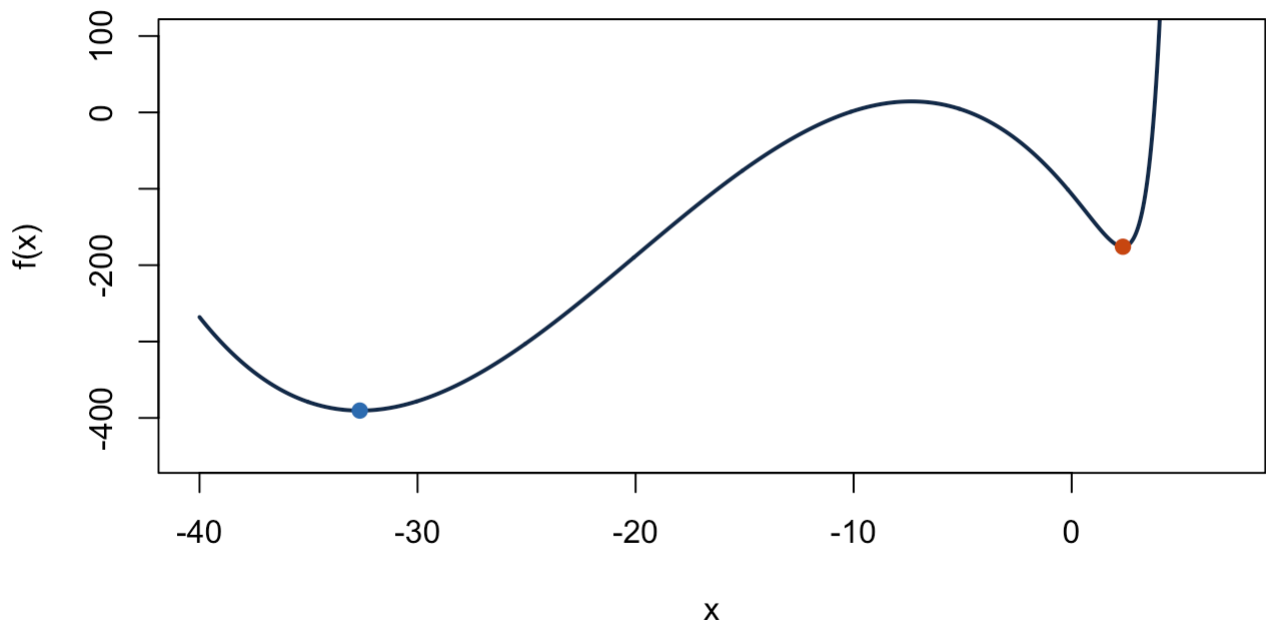
Likelihood profile for the coefficient of x2.

sigma2_mle	sigma2_unbiased	maximizing_t
0.4574759	0.4866765	0.0000000

Question 3 (Starting values in nonconvex optimization)

R

► Show the solution code



Nonconvex objective and the two BFGS solutions.

► Show the solution code

	starting_value	final_x	objective	absolute_gradient	converged
1	-15	-32.649111	-390.3858	8.725087e-08	TRUE
2	0	2.349967	-175.8626	6.556057e-07	TRUE

Question 4 (Nearly collinear predictors)

R

► Show the solution code

```

design condition_number training_rmse
1      X           1.524914      0.6763696
2 X_plus    20209.203501      0.6754668
  response beta_x1  beta_x6
1      y 351.6766 -350.2588
2 y_star 341.6766 -340.2588
  x1  x6
-10  10
fitted_difference_rmse fitted_difference_max
      0.001015879      0.002572675

```

Question 5 (Data preparation and summary statistics)

Solution

R

► Show the solution code

```
[1] 24 5
observation_id      group      x1      x2      response
"character"        "character"  "numeric"  "numeric"  "numeric"
duplicated_identifiers
0

observation_id      group      x1      x2      response
0                  0          0          0          3
group observations mean_response mean_feature_sum
A      A          7      5.600000      3.600000
B      B          7      7.602857      4.928571
C      C          7      7.997143      6.585714
observation_id group response feature_sum
22      obs-22      C      9.24      6.6
16      obs-16      B      9.00      5.2
14      obs-14      B      8.66      5.0
```

Question 6 (Can 39 million Fitbit records represent US adults?)

Solution

The 39 million daily records are repeated measurements from about 59,000 participants. Days from the same participant are generally correlated because health, occupation, habits, and environment persist over time. They should not be treated as 39 million independently sampled adults. An analysis of uncertainty must recognize participants as the sampling units or otherwise model the dependence among days from the same participant.

Participants also contribute different numbers of recorded days. Longer enrollment, consistent device use, regular synchronization, technical access, and fewer missing or invalid days can all increase the number of records contributed. Illness, travel, irregular work schedules, privacy concerns, device failure, or inconsistent use can reduce it. Averaging all recorded days gives more weight to participants with more recorded days. If the number of recorded days is related to activity, this record-weighted average can differ systematically from an average that gives each participant equal weight.

The comparison between BYOD and WEAR participants is not a randomized comparison of Fitbit ownership. People who already own a Fitbit may differ from people who enter through the free-device program in income, access to technology, health awareness, motivation, baseline activity, or other factors. The reported demographic differences provide direct evidence that the two routes contain different participant populations. Therefore, the difference in median steps could reflect selection into the two groups rather than a causal effect of already owning a Fitbit.

A US adult must pass through several stages before contributing an observed step count:

1. **Enrollment in All of Us.** Participation is voluntary. Enrollees may differ from other US adults in health, access to health care, interest in research, or willingness to share data.
2. **Entry through BYOD or WEAR.** A BYOD participant must already own a compatible Fitbit. A WEAR participant must be invited and agree to receive and use a device.

3. **Consent and technical connection.** A participant must agree to share wearable data and successfully connect an account.
4. **Continued device use and valid records.** The participant must continue to wear and synchronize the device, and each recorded day must satisfy the study's validity requirements.

Selection at any of these stages can affect the estimated population mean when inclusion is related to activity. A large number of records can reduce random variation around the mean for the observed participant-days, but it does not remove a systematic difference between those observations and the target population.

The naive average of all recorded daily step counts therefore has two problems for the stated target. First, it gives more weight to participants with more recorded days. Second, the observed participants may not represent US adults, even after each observed participant is given equal weight. Providing free Fitbits through WEAR reduces the barrier associated with already owning and purchasing a device, but it does not remove selection from enrollment, invitation and acceptance, data-sharing consent, account connection, continued use, or missing days.

An analysis aimed at the stated target should first summarize activity within each participant, account for dependence among repeated days, and then consider population weighting or calibration using information related to participation and activity. These adjustments require assumptions and cannot automatically remove selection caused by unmeasured factors.